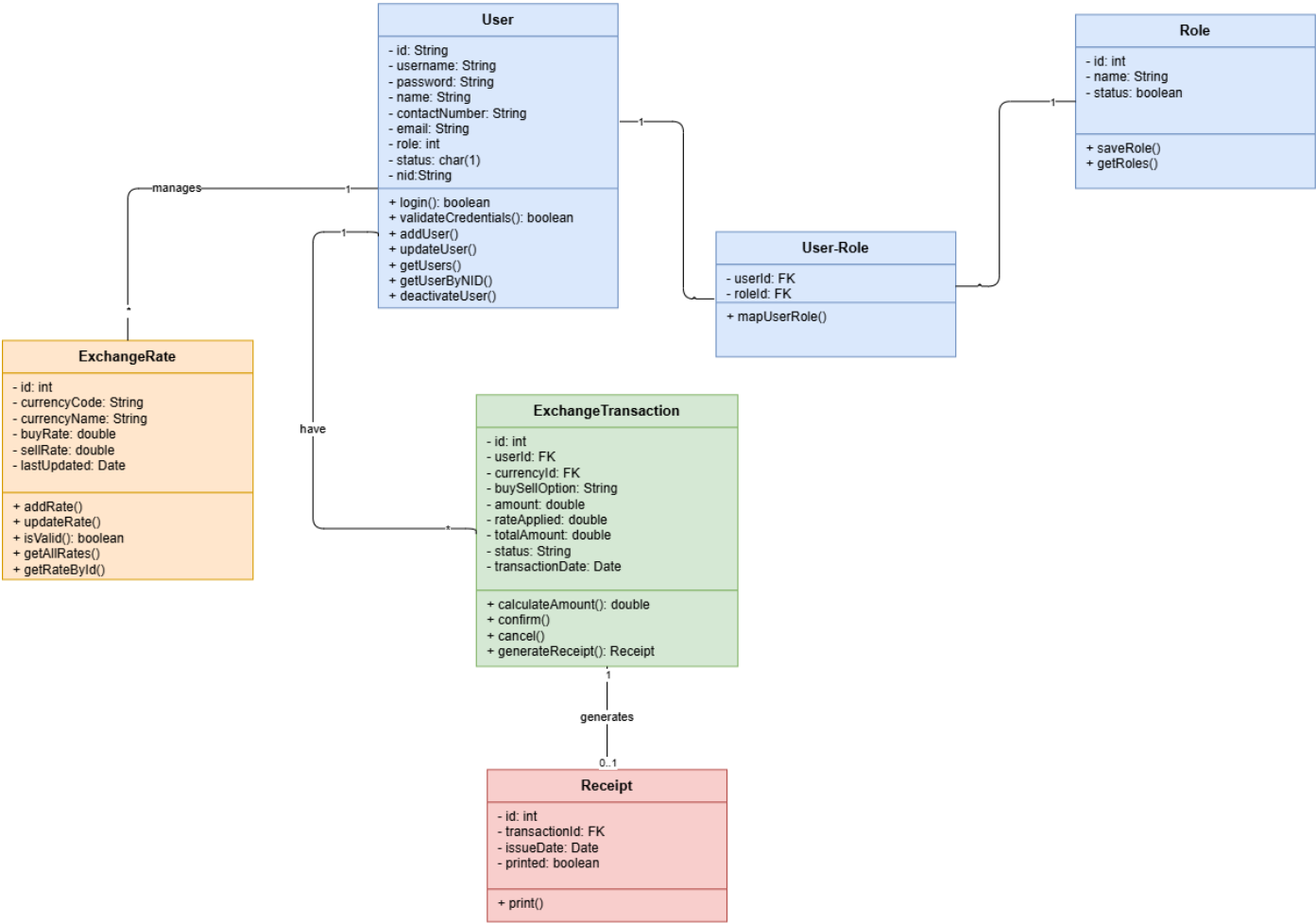


Money Exchange Management System — Class Diagram

The document shows the class diagram for money exchange rate system.

Class Diagram



Class classification

Class	Attributes	Methods	Description
User	id: String username: String password: String name: String contactNumber: String email: String status: char(1) nid: String	login(): boolean validateCredentials(): boolean addUser() updateUser() deactivateUser() getUsers() getUserByNID()	Define user information for both admin and customer.
Role	id: int name: String status: boolean	saveRole() getRoles()	Defines a role a user can hold (e.g. Customer, Admin) and its active status.

Class	Attributes	Methods	Description
User-Role	userId: FK roleId: FK	mapUserRole()	Shared class where user will mapped with the role
ExchangeRate	id: int currencyCode: String currencyName: String buyRate: double sellRate: double lastUpdated: Date	addRate() updateRate() isValid(): boolean getAllRates() getRateById()	Define buy/sell rate for a currency which will be maintained by an admin.
ExchangeTransaction	id: int userId: FK currencyId: FK buySellOption: String amount: double rateApplied: double totalAmount: double status: String transactionDate: Date	calculateAmount(): double confirm() cancel() generateReceipt(): Receipt	A single buy/sell request submitted by a customer against a given exchange rate.
Receipt	id: int transactionId: FK issueDate: Date printed: boolean	print()	Generated for a completed transaction. It can optionally be printed.

Relationships

- User have roles (many to many) — a user can hold multiple roles, and a role can apply to many users.
- User manages ExchangeRate (1 to many) — 1 (admin) to many (rates).
- User have ExchangeTransaction (1 to many) — 1 (customer) to many (transactions).
- ExchangeTransaction generates Receipt (1 to 0/1) — 1 to 0..1 (a receipt only exists if requested/printed).